The True Cost of a Warming Climate: Infrastructure Collapse and Coastal Loss

1. The "Sinking Nations" Misconception

When analyzing the catastrophic environmental risks projected for 2100, it is critical to dispel a widespread misconception: the idea that rising sea levels will swallow entire countries whole. With the sole exception of extremely low-lying island archipelagos like the Maldives, no sovereign nation is at risk of being completely submerged. The true, imminent danger of the definitive warming trendline—which shifted historical global land temperatures from a ~8.0°C baseline to over 9.5°C—is far more insidious than countries simply vanishing beneath the waves.

2. The Project's Core Conclusion: The Mechanics of Coastal Failure

The primary conclusion drawn from projecting historical temperature acceleration into future inundation models is that nations will be crippled from the inside out, long before the land itself is permanently underwater. The genuine existential threats by 2100 are driven by three interconnected, systemic failures:

- **Systemic Infrastructure Failure:**

  This is the most critical and immediate threat. Encroaching tides and rising water tables will utterly overwhelm urban lifelines. Municipal sewage and water management systems will fail, causing drainage networks to back up. As floodwaters mix with compromised sanitation systems, fresh water supplies will be severely contaminated, rendering coastal cities uninhabitable due to disease and structural decay.

- **Exponentially Increased Flooding:** A rising ocean baseline exponentially amplifies the destructive power of natural weather volatility. Storm surges and seasonal monsoons will penetrate significantly further inland. What were historically rare, manageable flood events will transform into chronic, devastating occurrences that constantly disrupt local economies and destroy housing.

- **Coastal Erosion and Permanent Land Loss:** As sea levels rise, the continuous kinetic energy of the ocean physically degrades coastlines. This leads to the permanent loss of vital agricultural lands through relentless erosion and toxic saltwater intrusion, crippling food supplies.

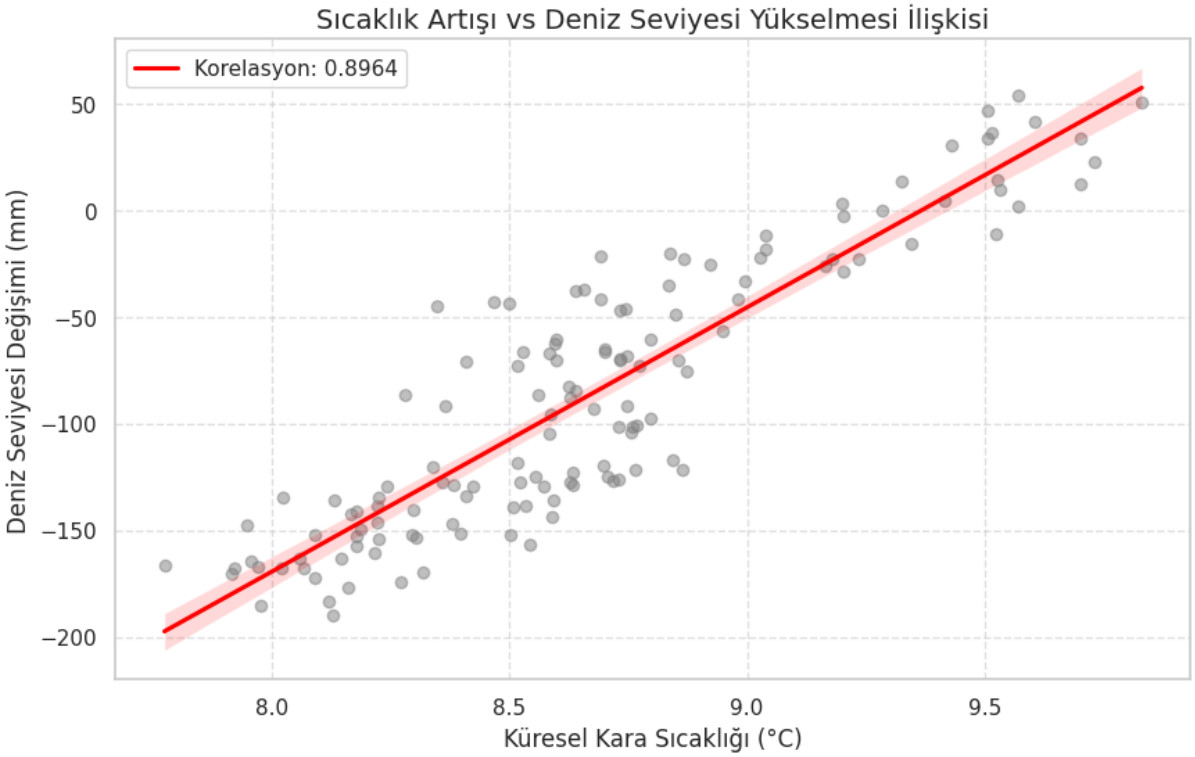
3. The Human Impact Multiplier

The severity of these projected crises is dictated by population density, not just geography. For example, while a nation like Bangladesh will not entirely disappear, its highly populated, low-lying delta regions will face unparalleled devastation. The combination of chronic inland flooding, irreversible coastal erosion, and the total collapse of sanitation infrastructure will render massive swathes of the country completely unlivable.

Ultimately, this project concludes that the true risk of 2100 is not disappearing landmasses, but the catastrophic, irreversible loss of coastal habitability and the subsequent displacement of hundreds of millions of people.

4. Quantitative Correlation: Linking Thermal Expansion to Sea Level Rise

Following the quantitative visualization of the robust relationship between the anomalies in global mean temperature and the rise in global mean sea level, as depicted in the preceding analysis, it becomes imperative to transition from historical observations to future projections. This sequential analysis seeks to transform the identified correlation into a robust predictive framework, moving beyond descriptive statistics toward a more sophisticated model evaluation and selection process.



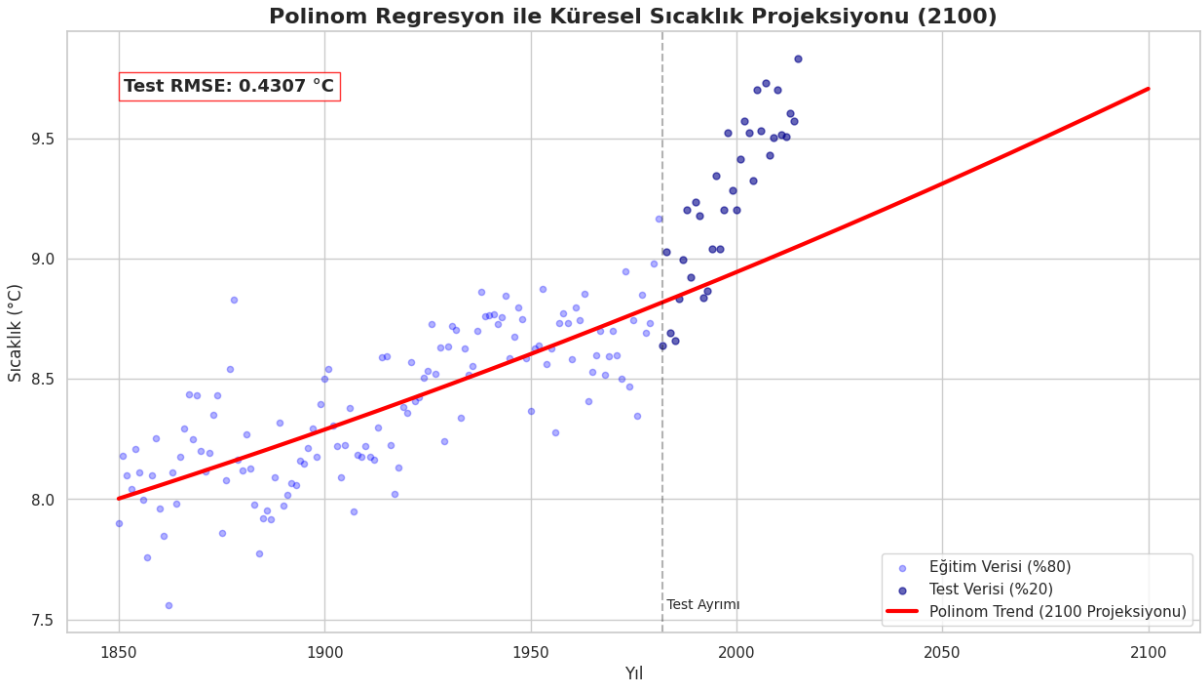
5. Predictive Framework: Comparative Analysis of Forecasting Models

The forecasting models developed within the scope of this project represent various learning paradigms utilized in time series analysis. Specifically, 2nd Degree Polynomial Regression was established to track non-linear trends, while XGBoost and Random Forest were implemented to leverage boosting-based error optimization and bagging-based variance reduction, respectively. Throughout the benchmarking process, the RMSE (Root Mean Square Error) metric was utilized due to its sensitivity to outliers and its direct unit-based interpretability. By employing this metric, the projection capacities of the models have been quantitatively demonstrated through numerical data.

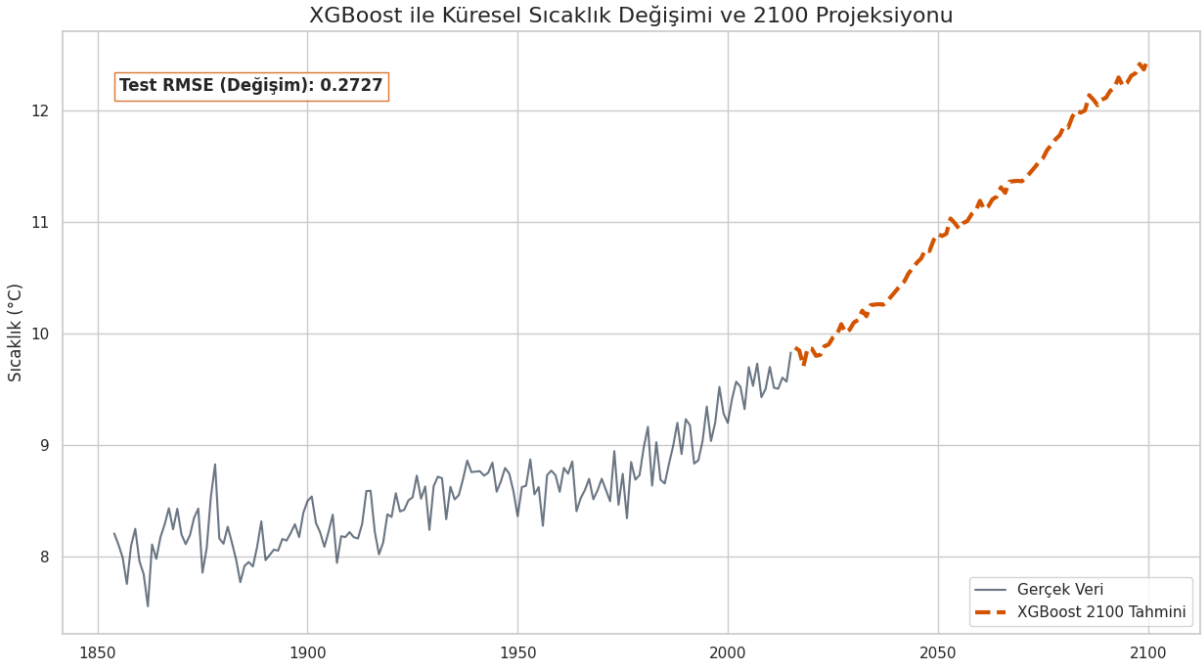
6. Model Benchmarking and Performance Evaluation

6.1 Global Temperature Increase Future Forecast

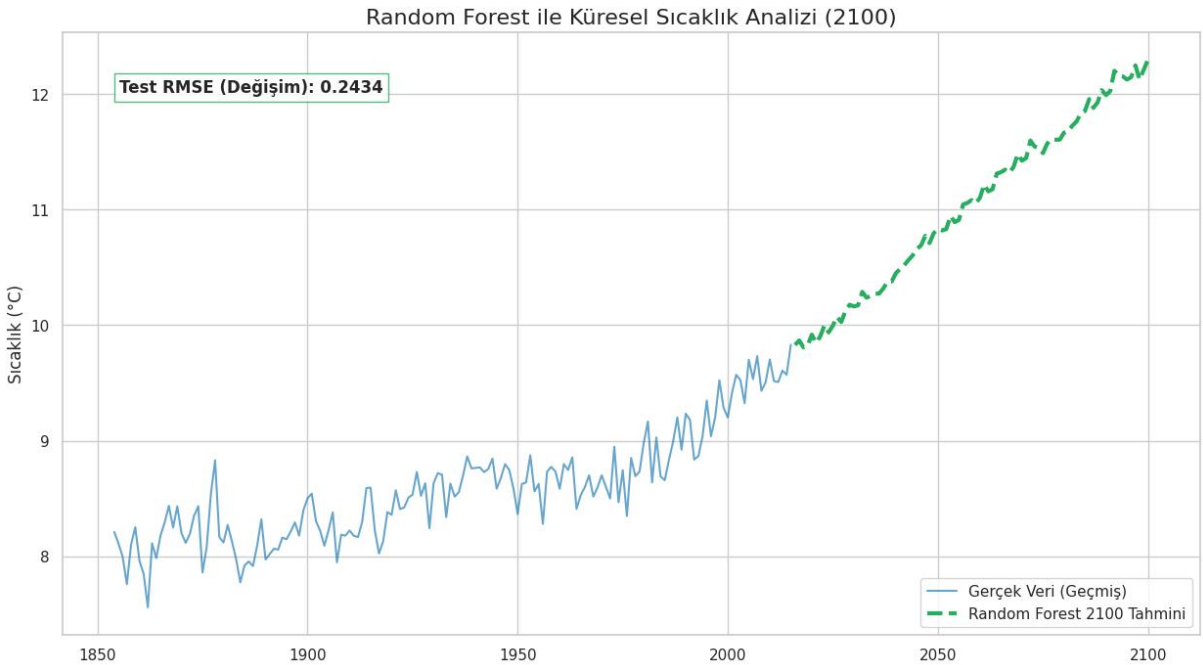
6.1.1 Polynomial Regression Global Temperature Increase Future Forecast



6.1.2 XGBoost Global Temperature Increase Future Forecast



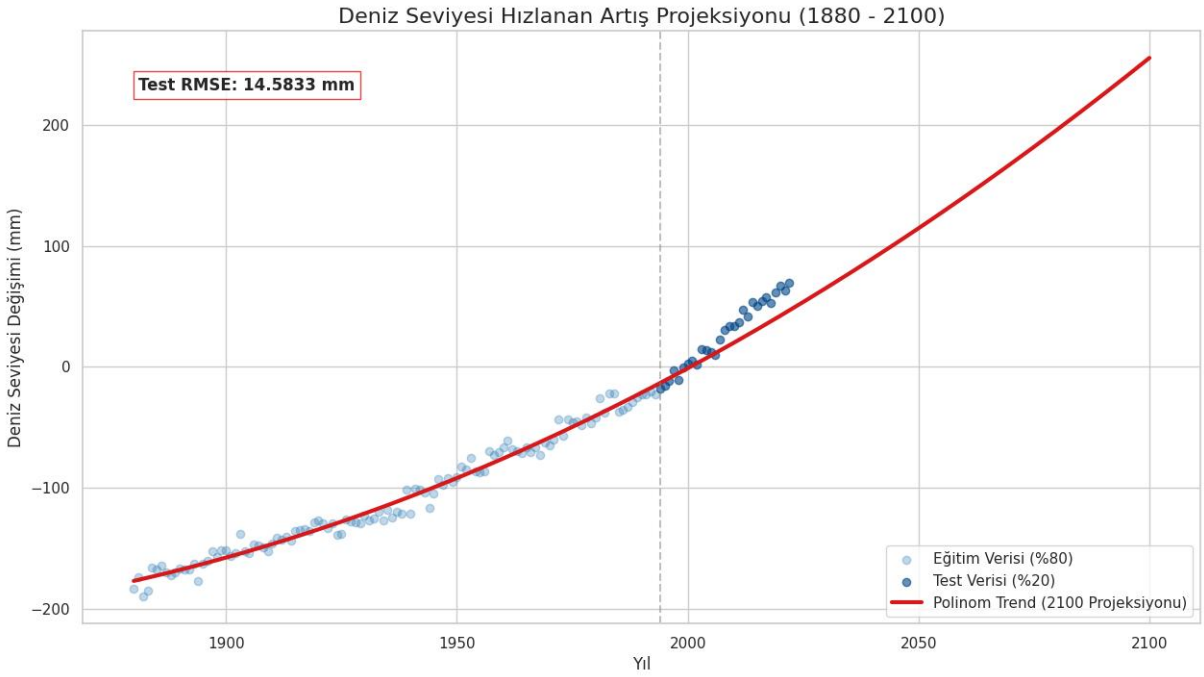
6.1.3 Random Forest Global Temperature Increase Future Forecast



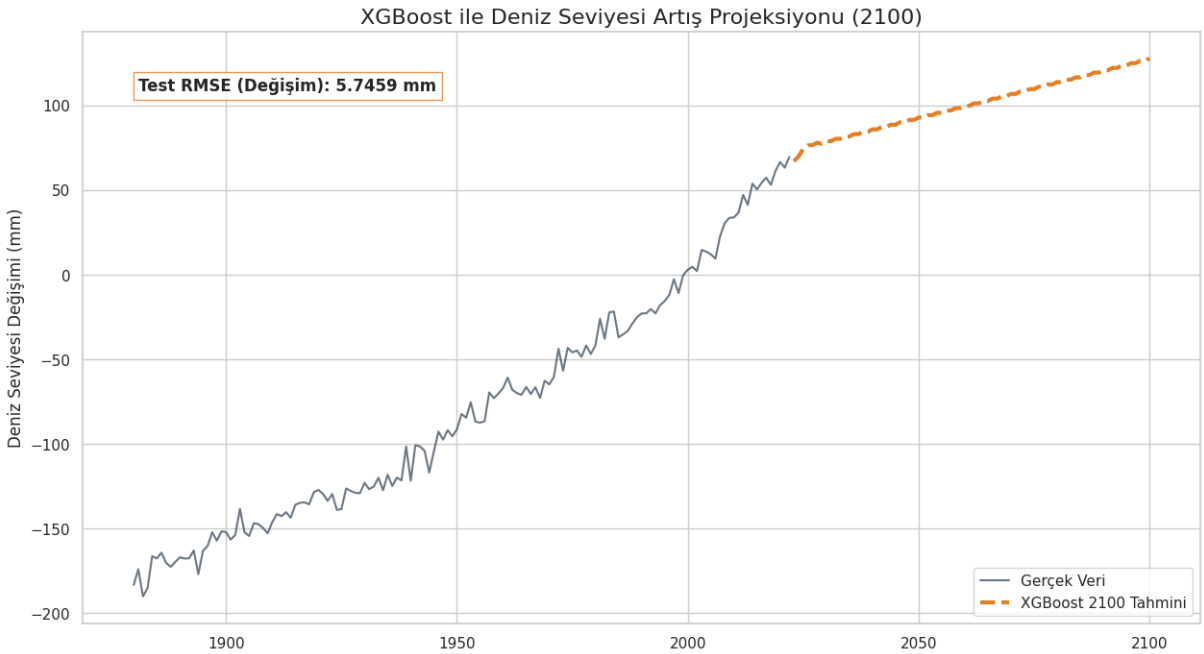
As evidenced by the empirical results, the Random Forest model emerged as the most accurate architecture for this climate projection. The model achieved an RMSE of 0.2434, indicating that its predictions deviate from the actual observed values by an average magnitude of only 0.2434 °C. This superior performance confirms that the Random Forest algorithm is the most reliable foundation for the 2100 projections.

6.2 Sea Level Rise Increase Forecast

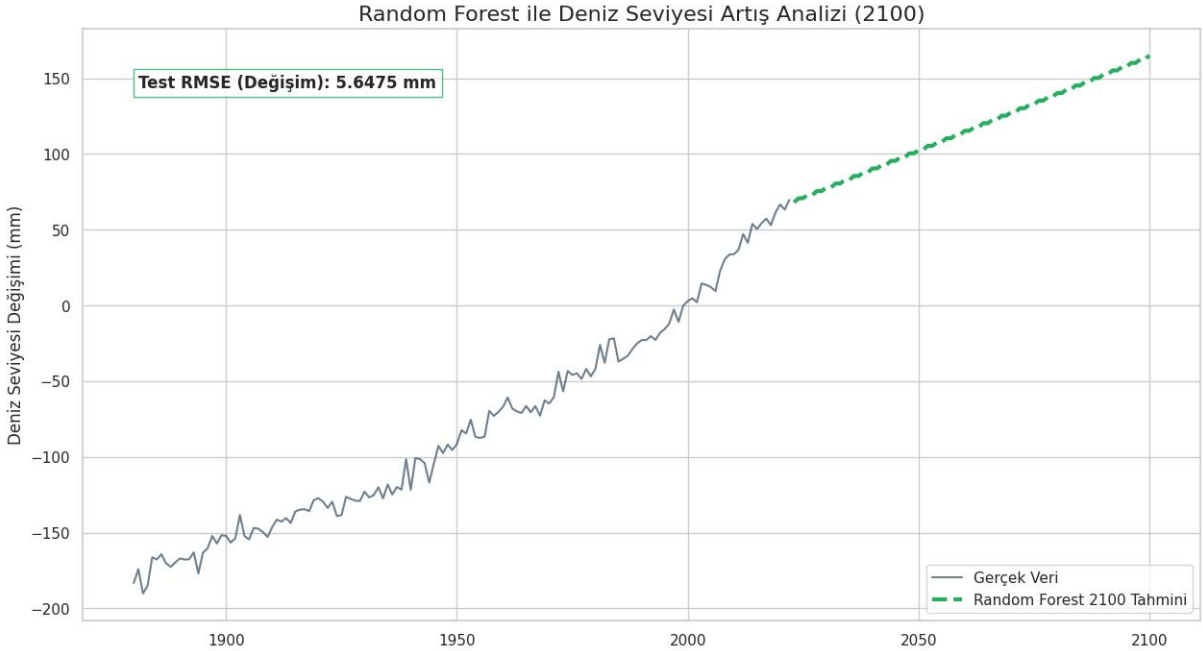
6.2.1 Polynomial Regression Sea Level Rise Increase Forecast



6.2.2 XGBoost Sea Level Rise Increase Forecast



6.2.3 Random Forest Sea Level Rise Increase Forecast



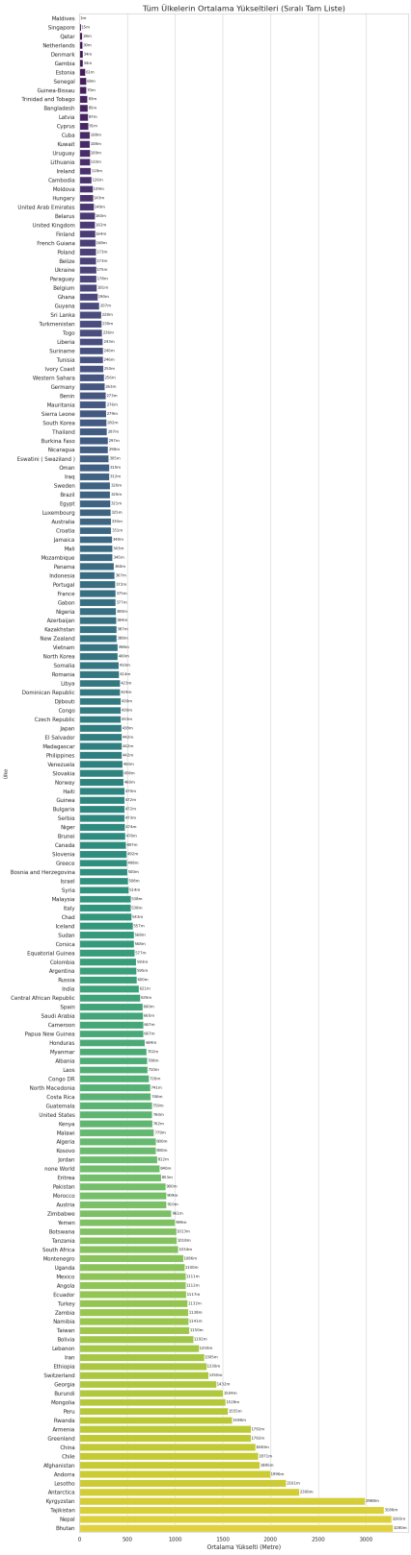
In the comparative assessment of sea level rise projections, the **Random Forest** model demonstrated the highest predictive precision, narrowly outperforming the highly competitive XGBoost architecture. The model achieved a remarkably low **RMSE of 5.6475 mm**, signifying that its projections deviate from historical observations by an average of just over half a centimeter. Considering the massive scale and inherent volatility of global oceanic data, this margin of error represents a high degree of calibration. The results confirm that the Random Forest algorithm effectively neutralizes stochastic fluctuations, providing a stable and robust quantitative basis for estimating inundation risks toward the year 2100.

7. Conclusion: The Quantitative Verdict on Systemic Coastal Failure

The rigorous comparative analysis conducted throughout this project transforms the abstract threat of climate change into a definitive, data-driven reality. By selecting the Random Forest model—validated by a remarkably low RMSE of 0.2434 °C for temperature and 5.64 mm for sea level—we have established a high-precision forecasting foundation that confirms an irreversible upward trajectory toward 2100.

As illustrated in the global distribution graph below, the impact is not uniform; however, the systemic risk is universal. The data proves that the 'Sinking Nations' myth oversimplifies a much more complex catastrophe. The real crisis, as quantified by our models, lies in the loss of habitability driven by infrastructure collapse, sanitation failure, and agricultural degradation. The 2100 projections indicate that even a few centimeters of deviation in sea level will be enough to trigger a domino effect across global supply chains and urban lifelines.

Ultimately, this project concludes that mitigating the true cost of a warming climate requires a shift from viewing sea level rise as a distant geographic event to recognizing it as an immediate threat to global systemic resilience. The mathematical certainty of these projections serves as a final call for adaptive engineering and infrastructure overhaul, before the cost of maintenance exceeds the possibility of survival.

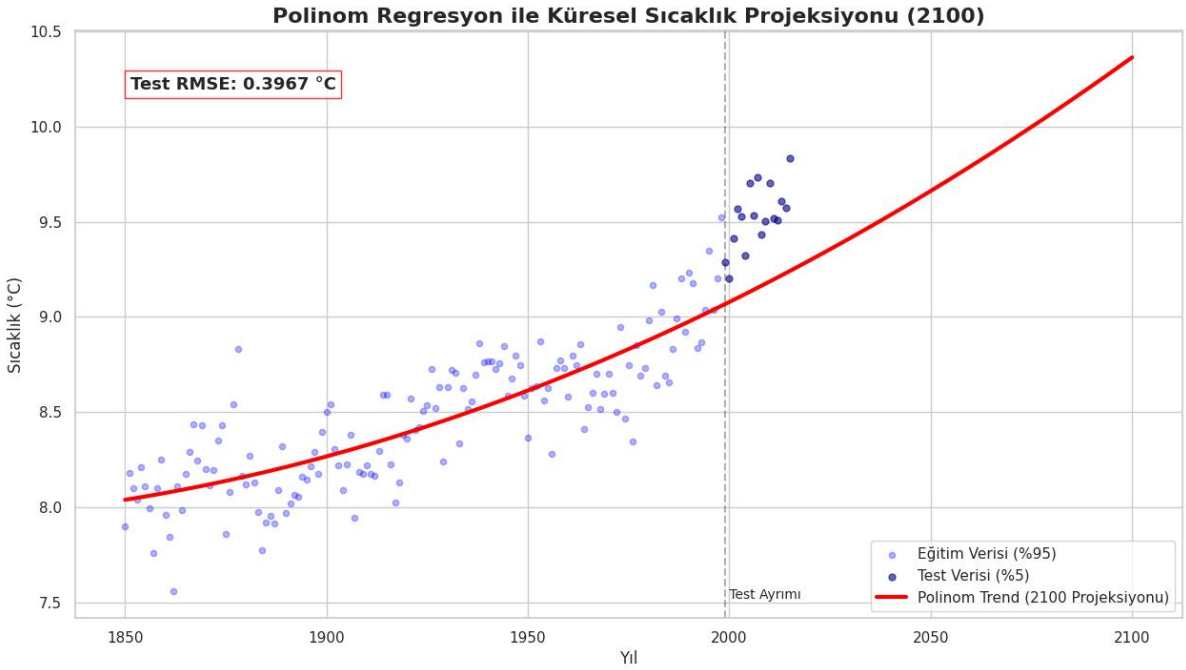


8. Rearrangement: Experiment On Hyperparameters

8.1 Global Temperature Increase Future Forecast:

8.1.1 Polynomial Regression:

- Degree: 2, Test size: 0.1



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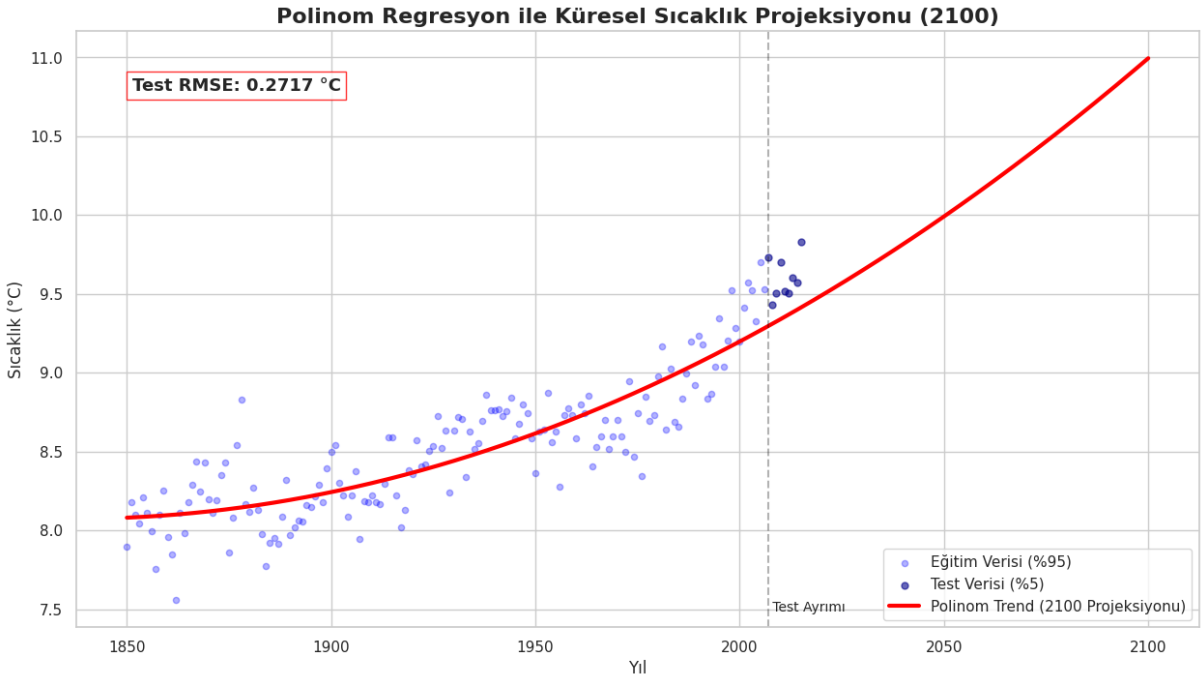
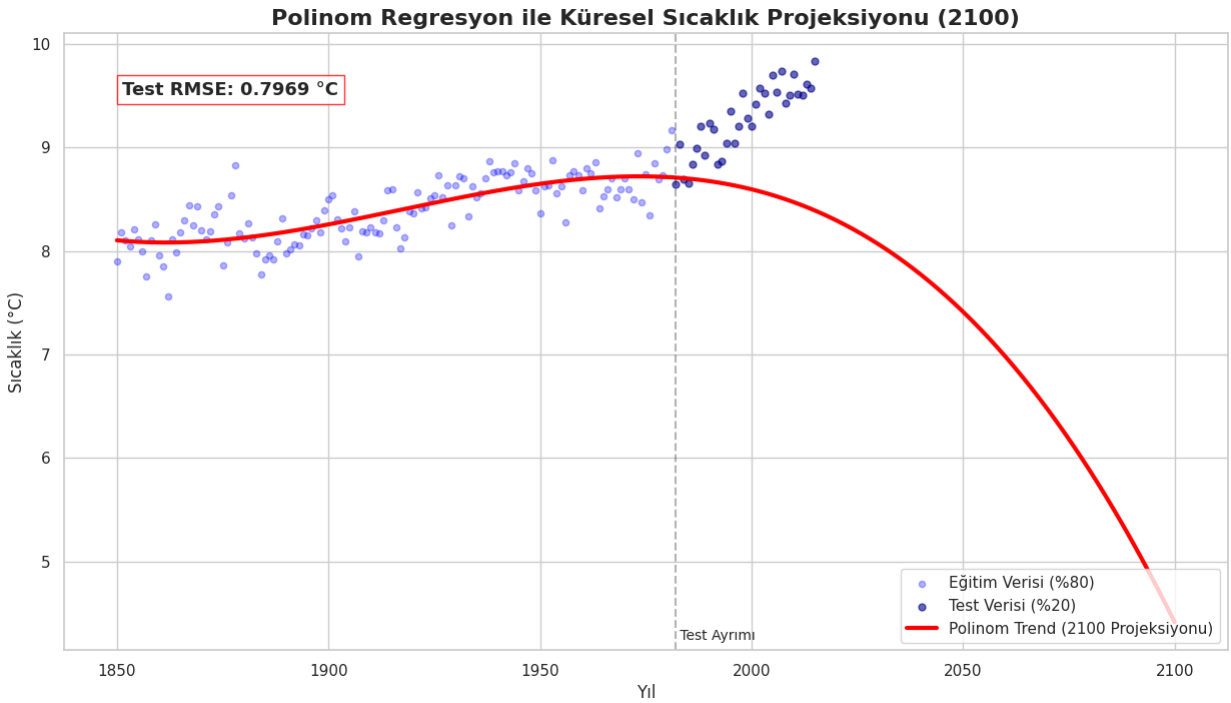


- Degree: 3, Test size: 0.2



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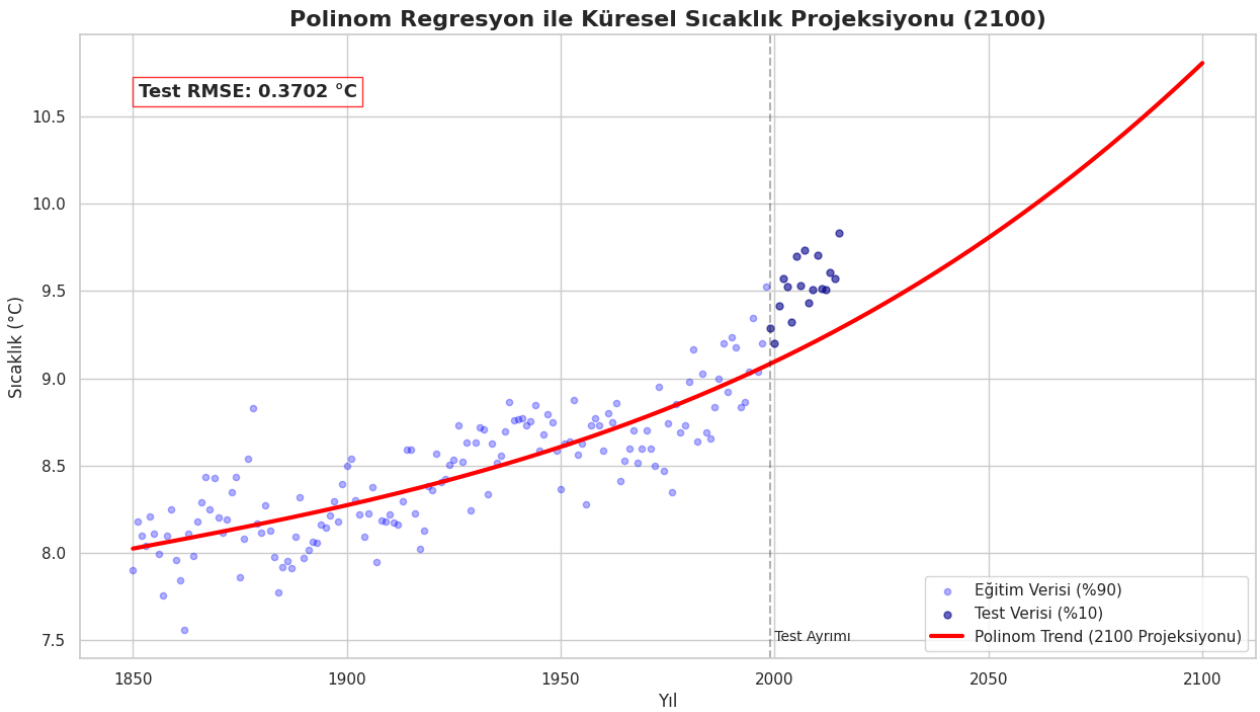

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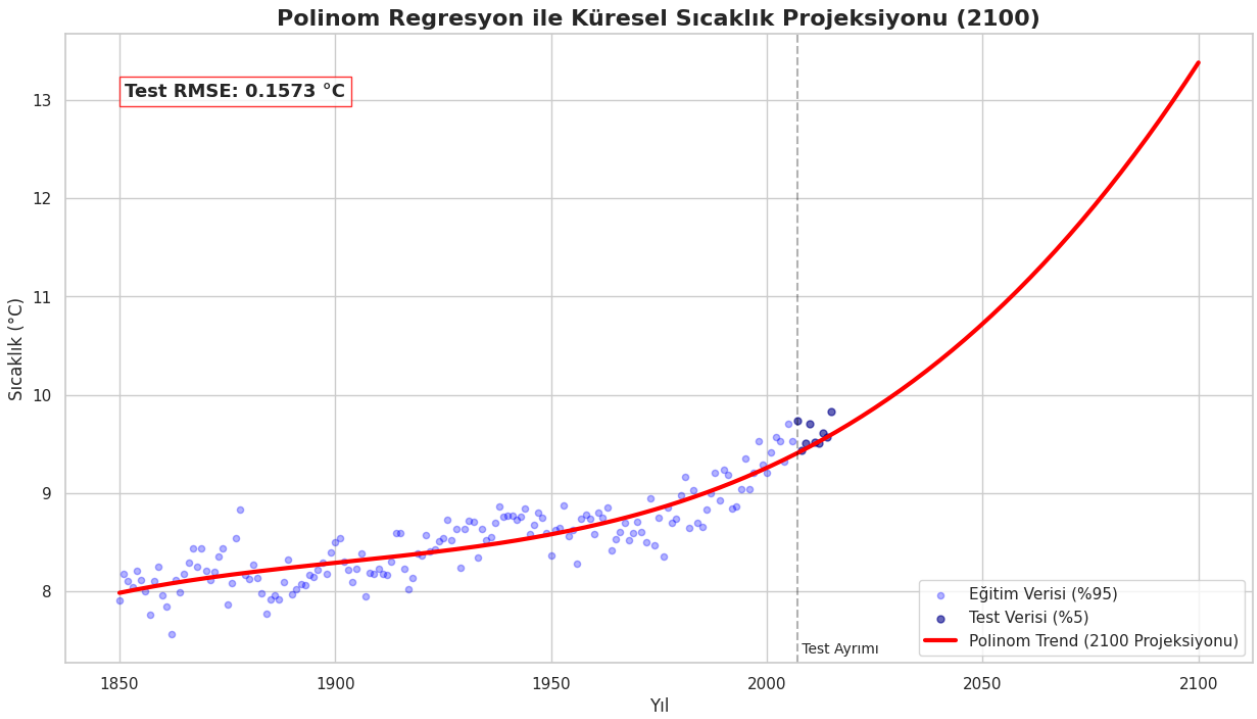
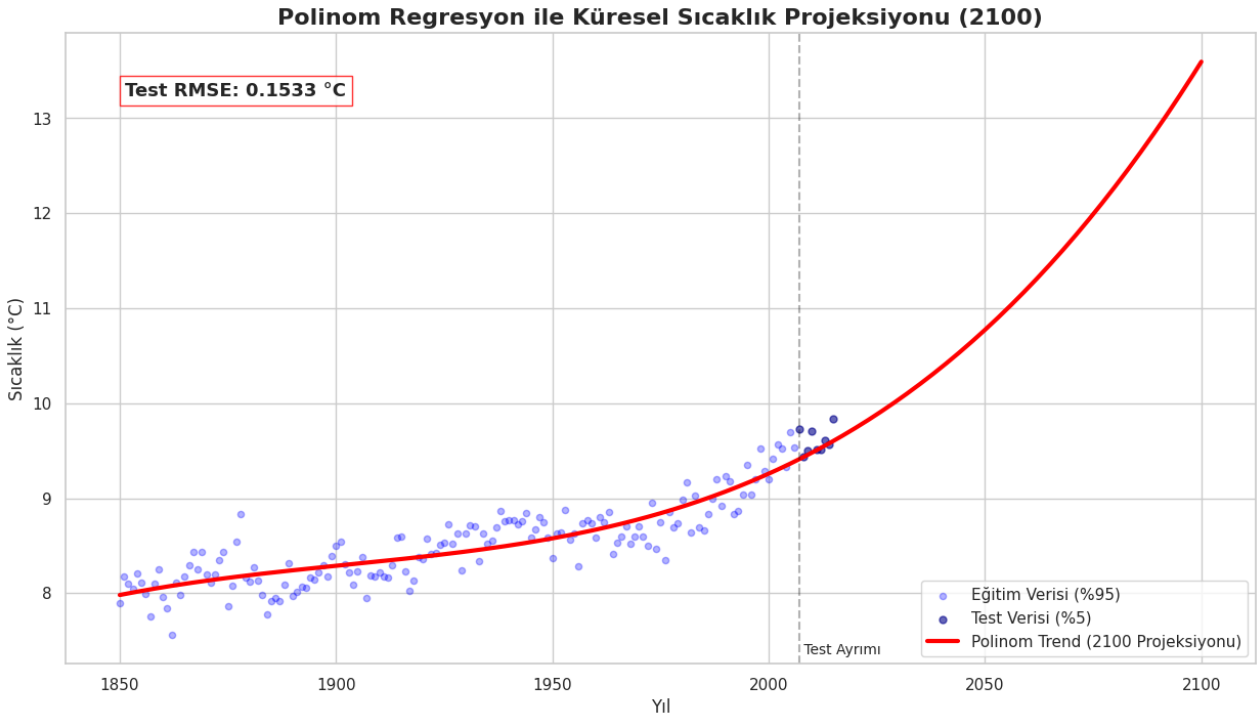
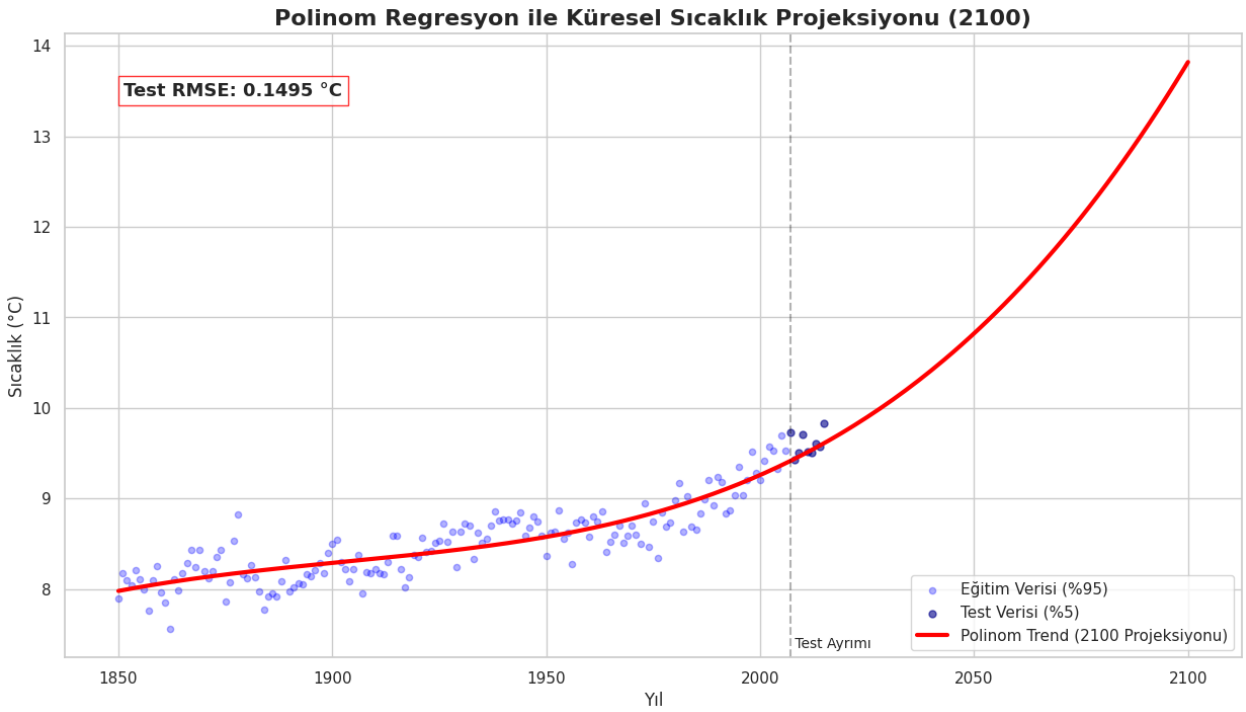


- Degree: 4, Test size: 0.05

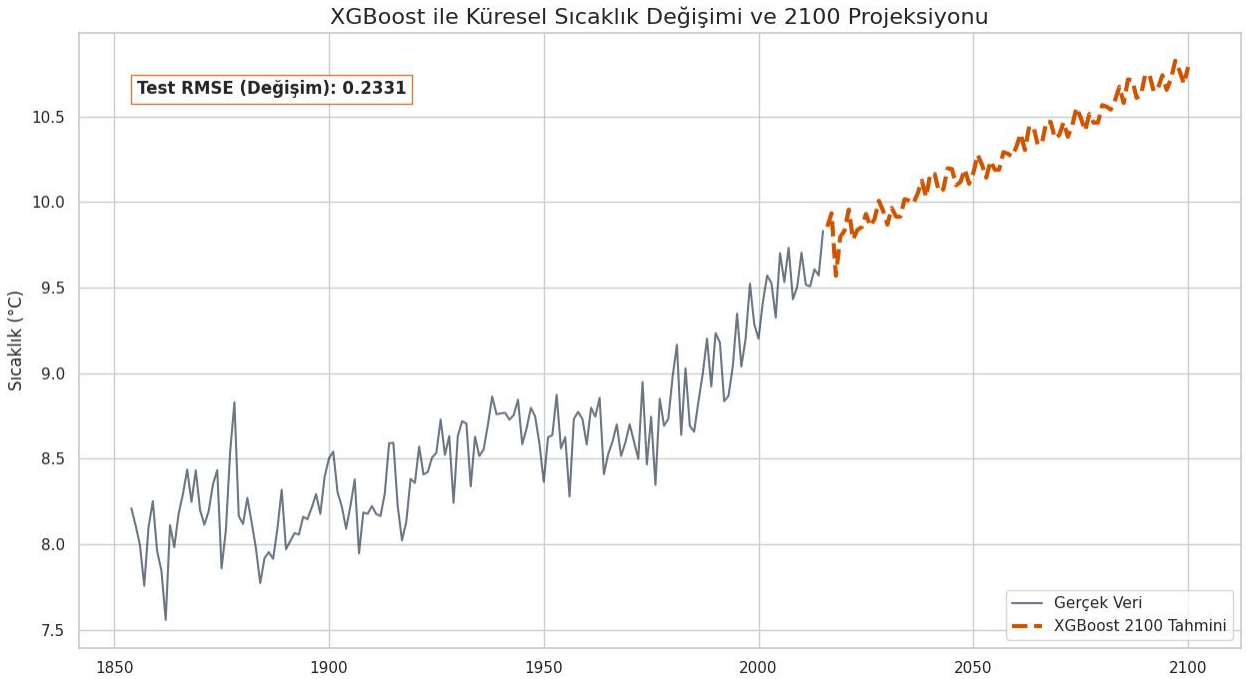


- Degree: 5, Test size: 0.05

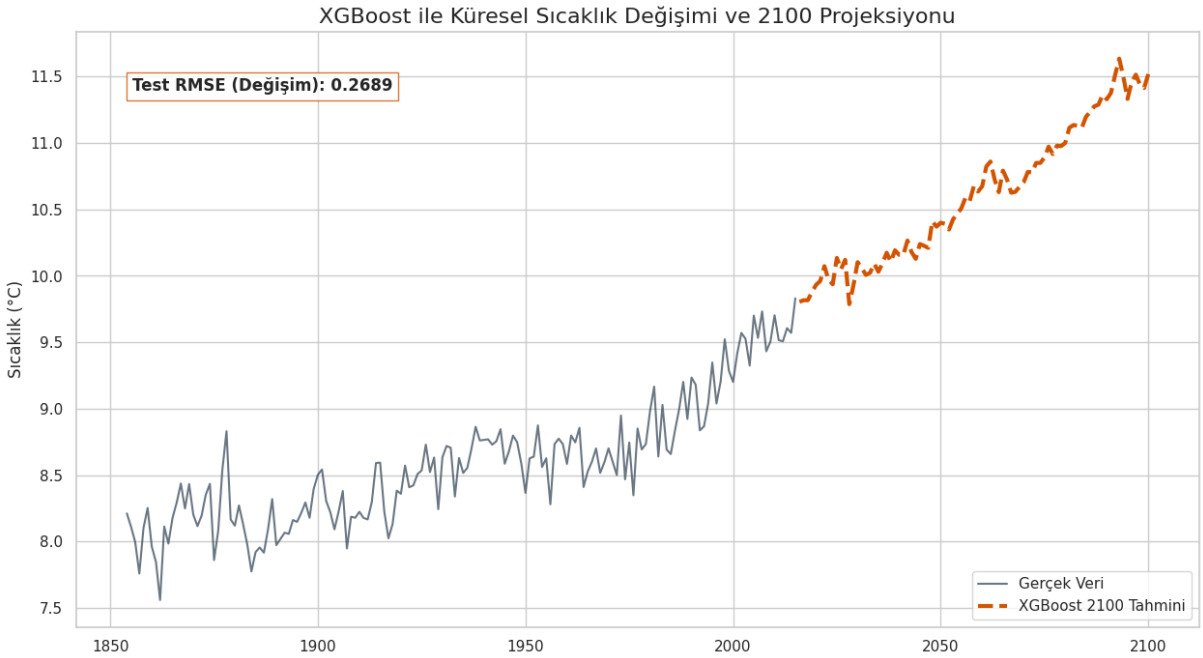


8.1.2 XGBoost:

- Test size: 0.1

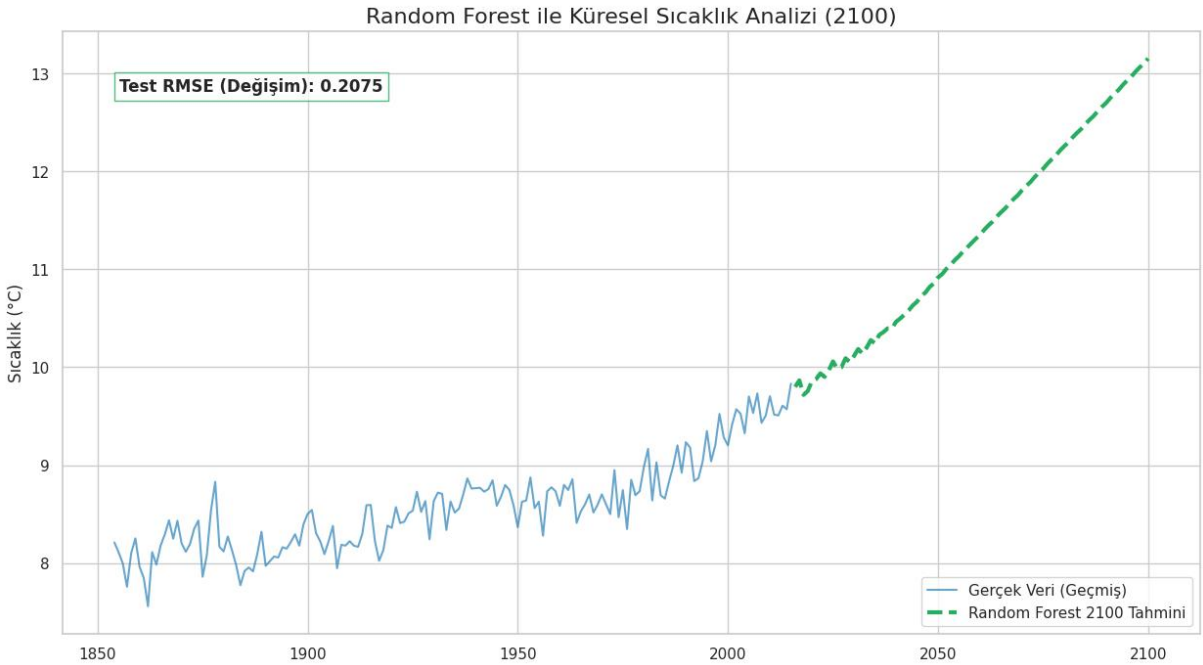


- Test Size: 0.05

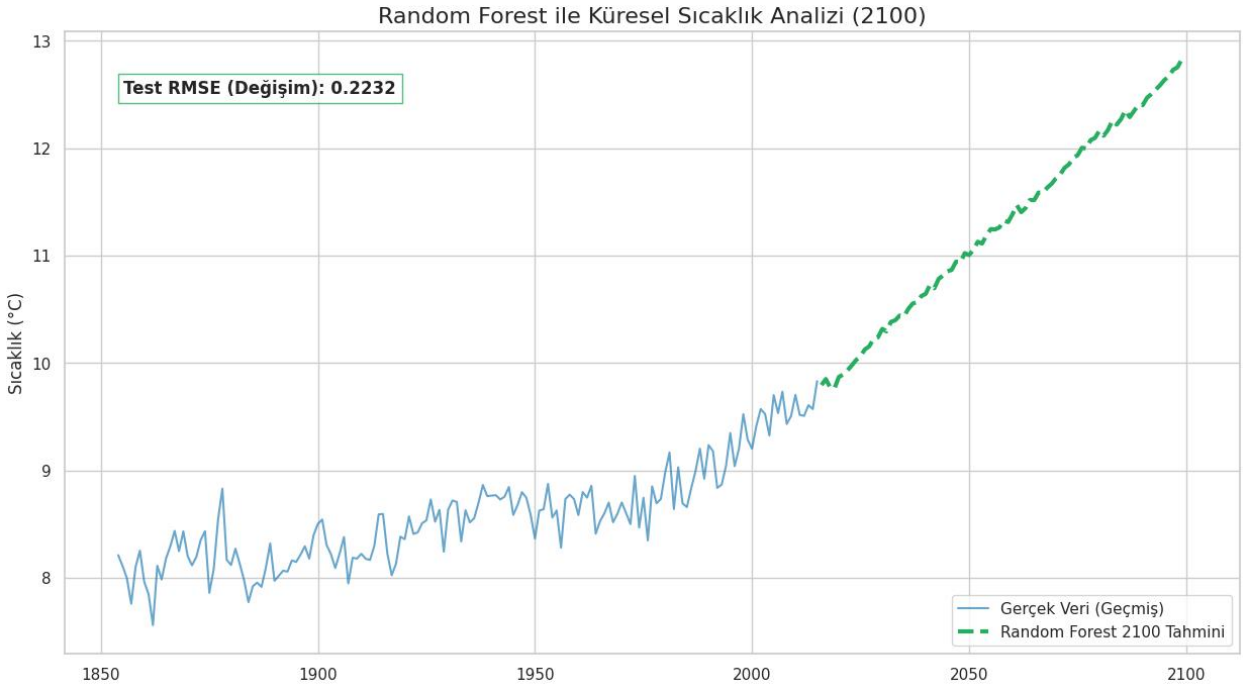


8.1.3 Random Forest:

- Test Size: 0.1



- Test size: 0.05



8.2 **Sea Level Rise Increase Forecast:**

8.2.1 Polynomial Regression:

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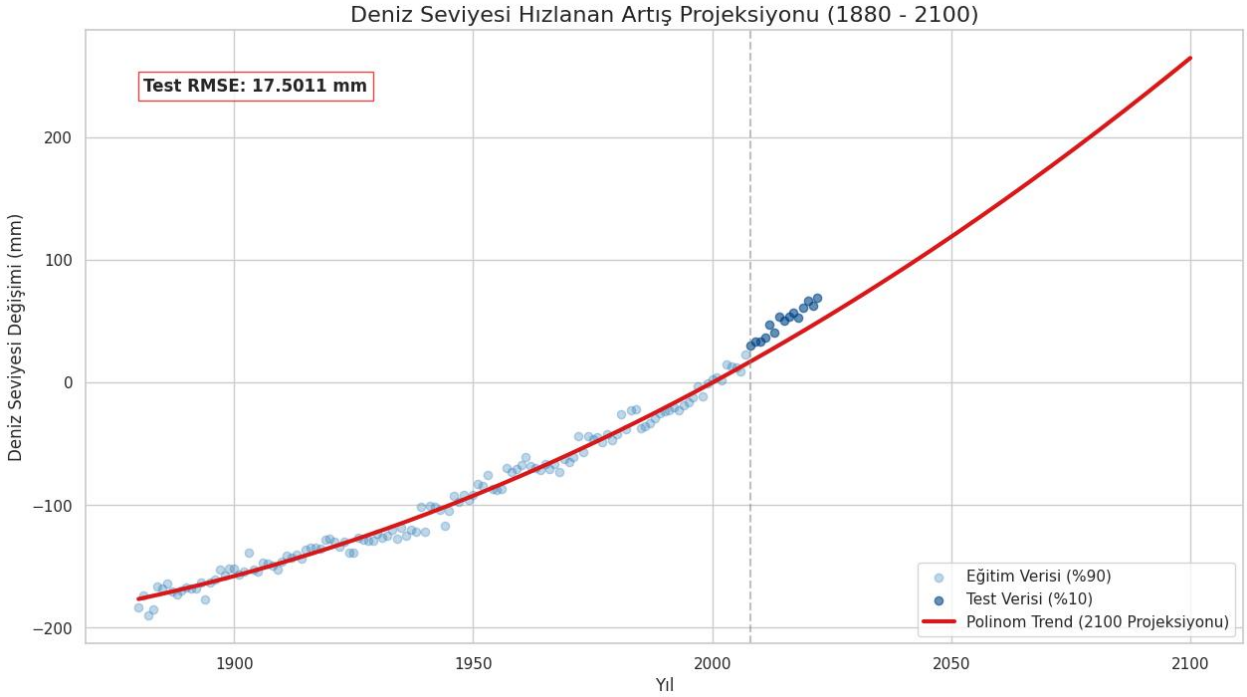
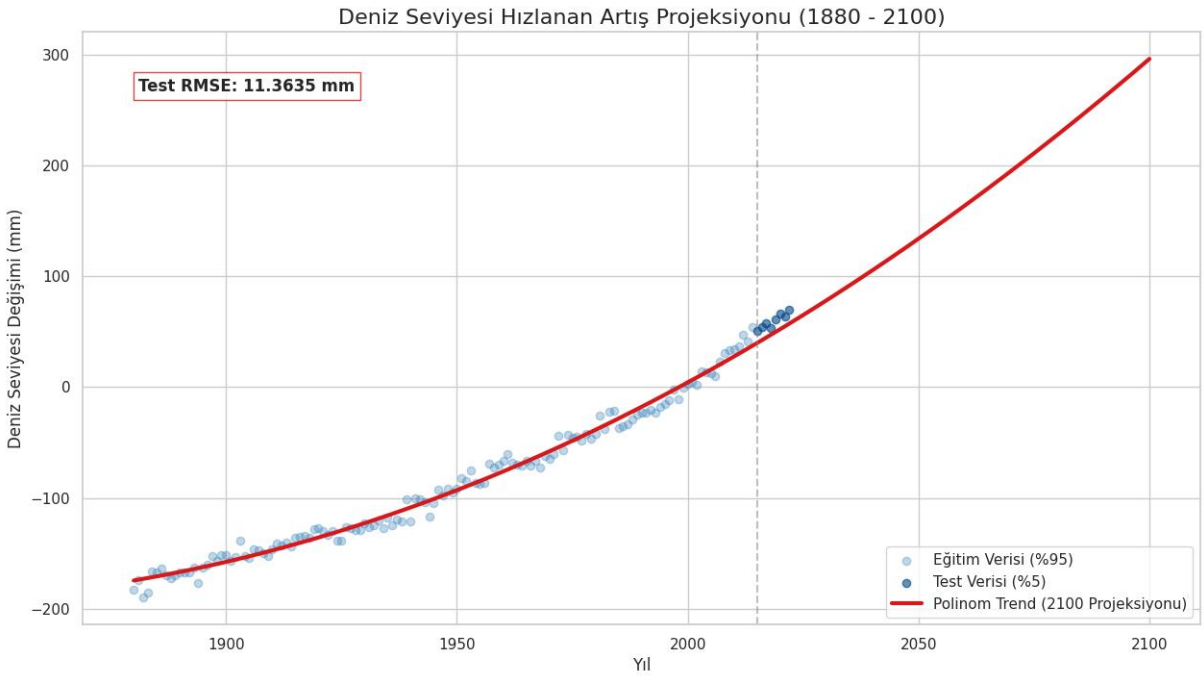
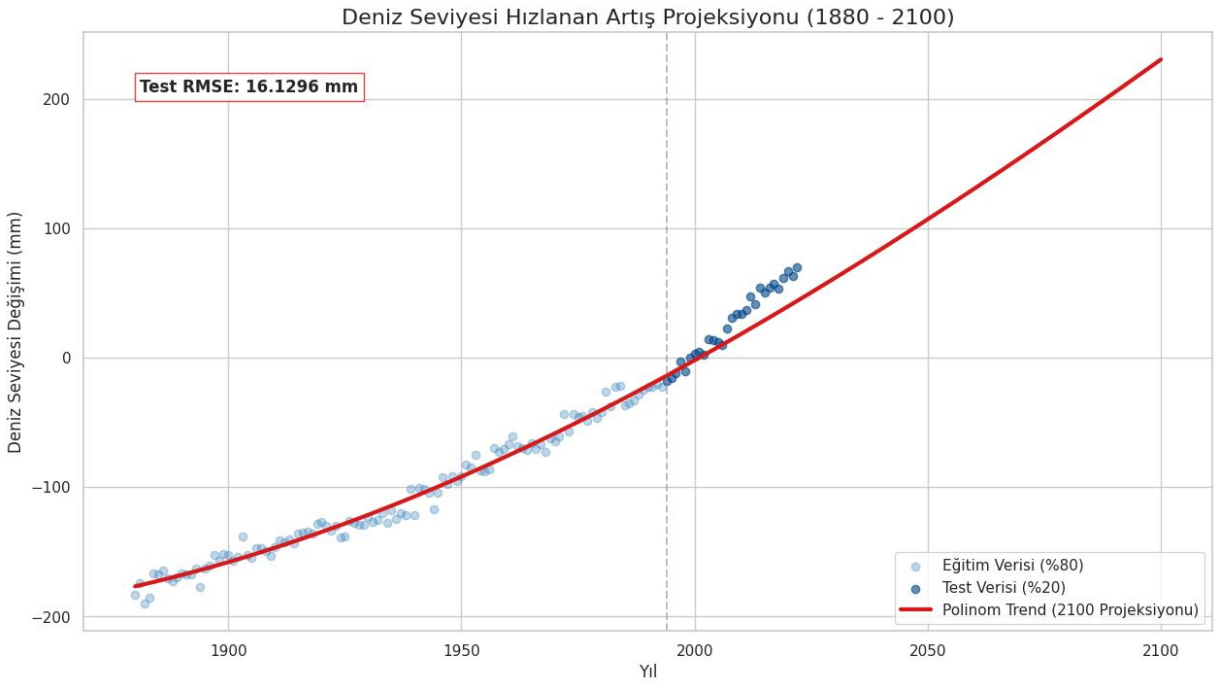


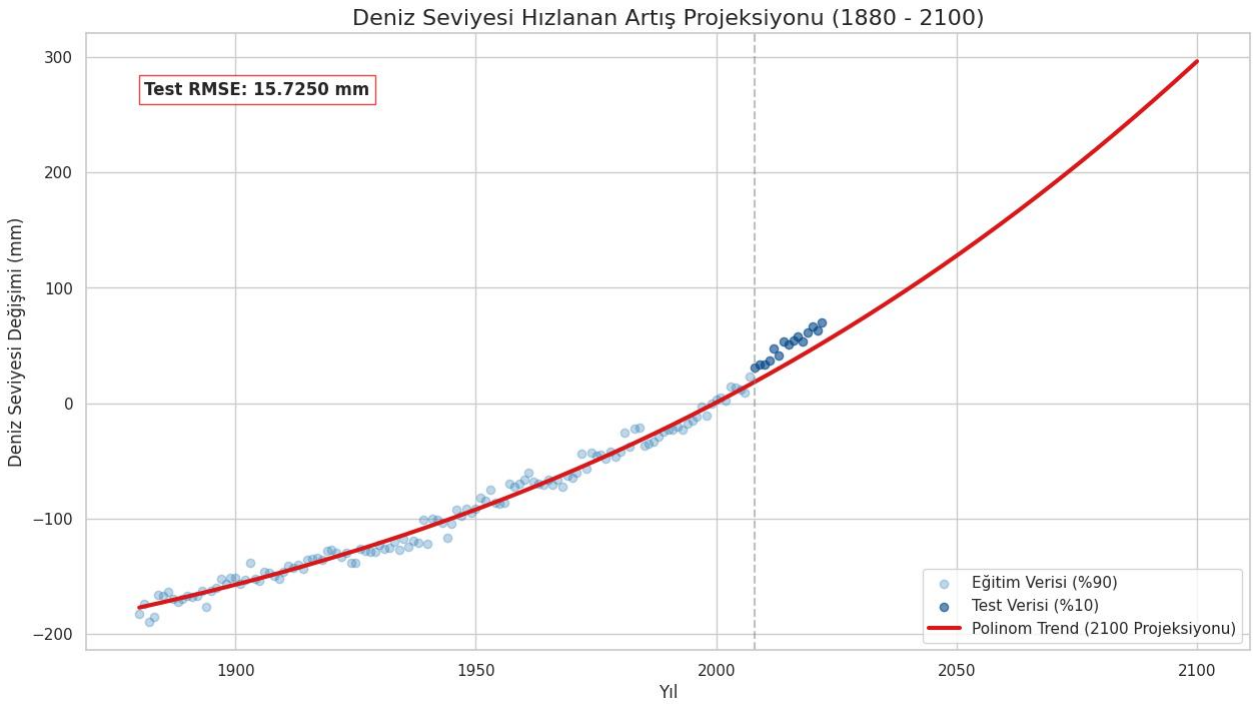
- Degree: 2, Test size: 0.05



- Degree: 3, Test size: 0.2



- Degree: 3, Test size: 0.1



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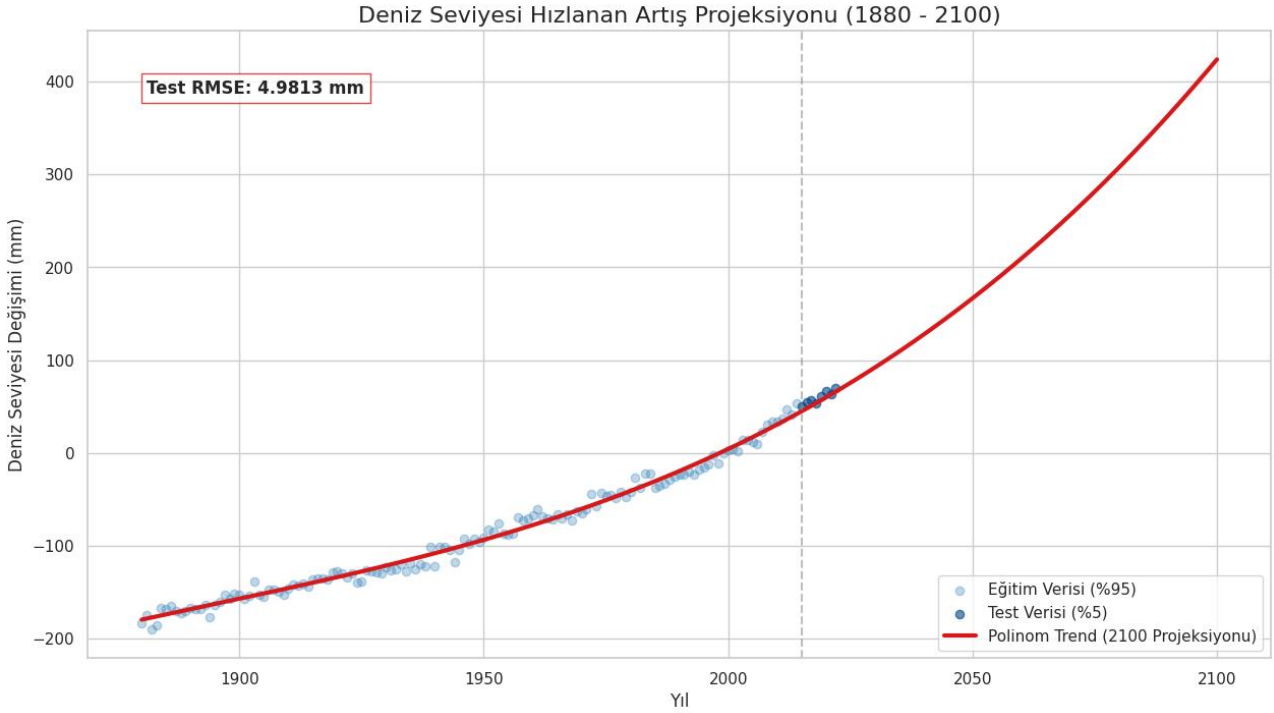
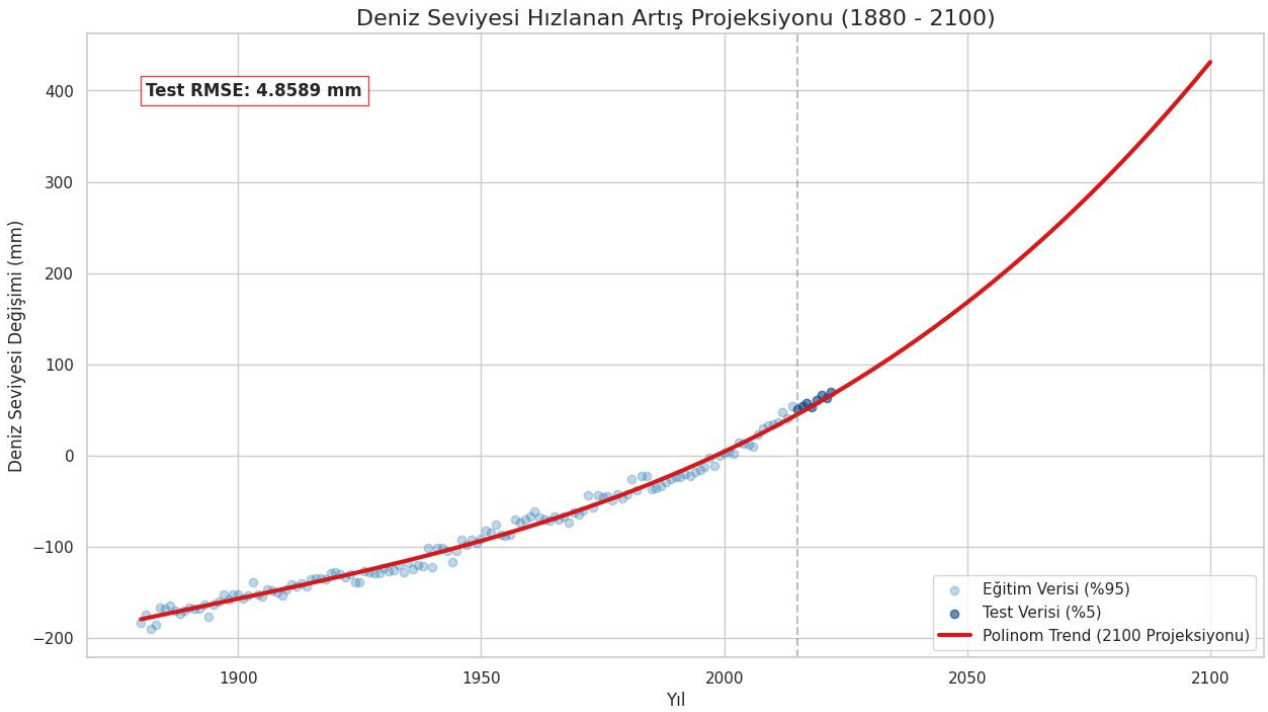
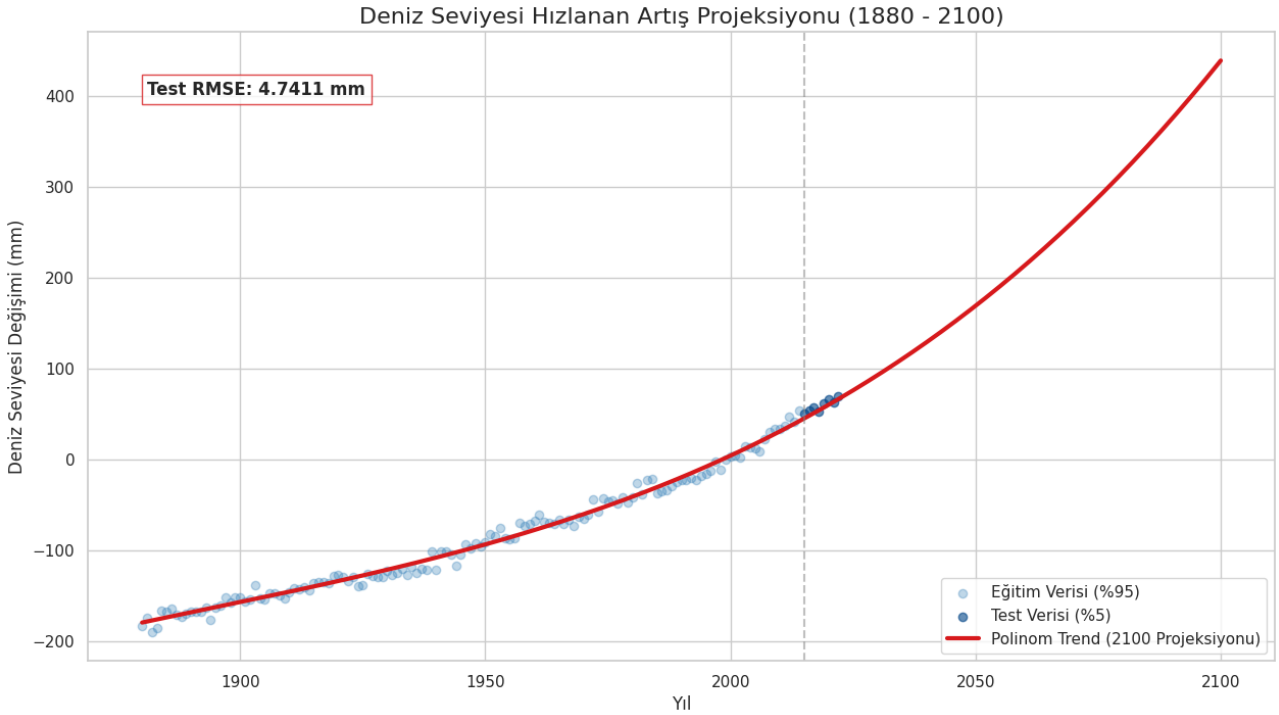


- Degree: 4, Test size: 0.05

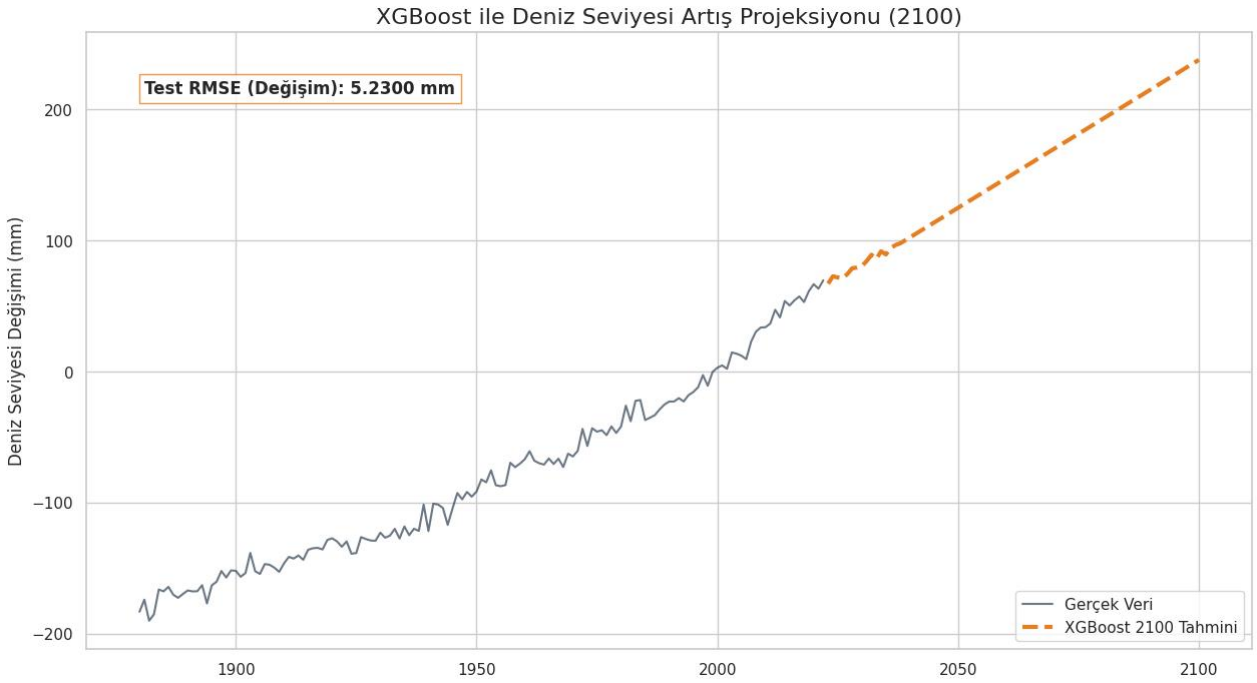


- Degree: 5, Test size: 0.05

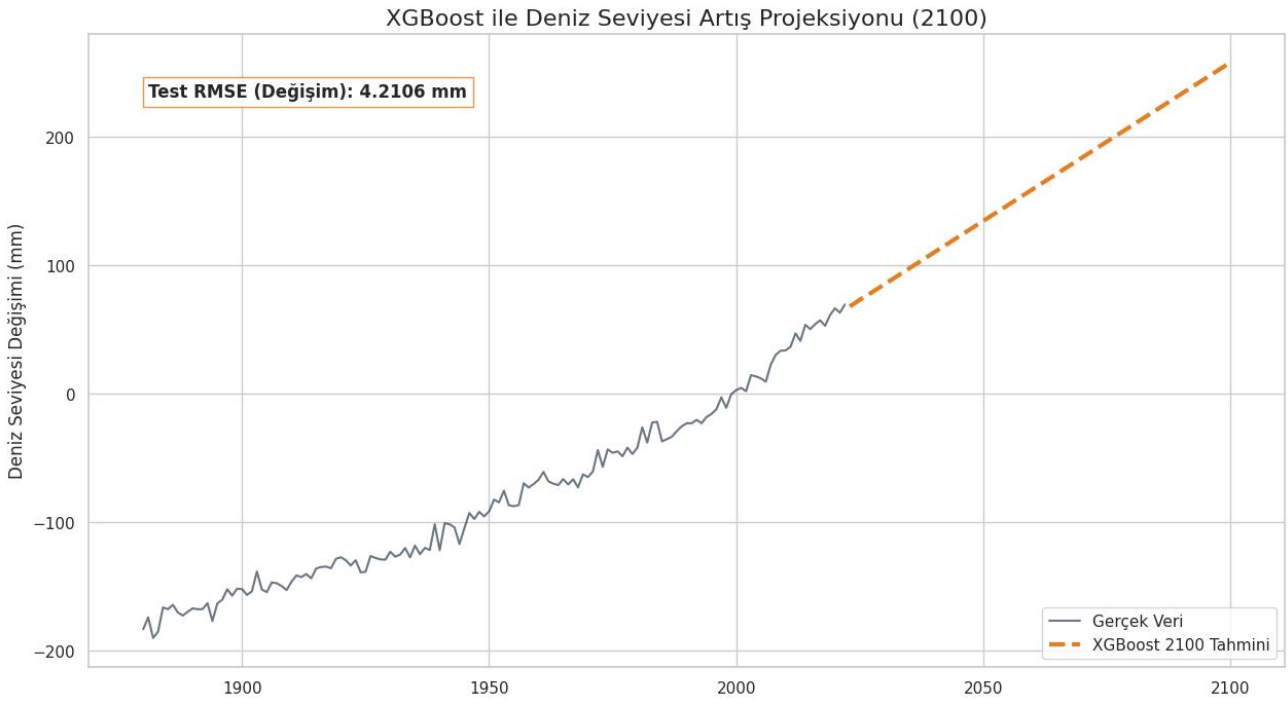


8.2.2 XGBoost:

- Test size: 0.1

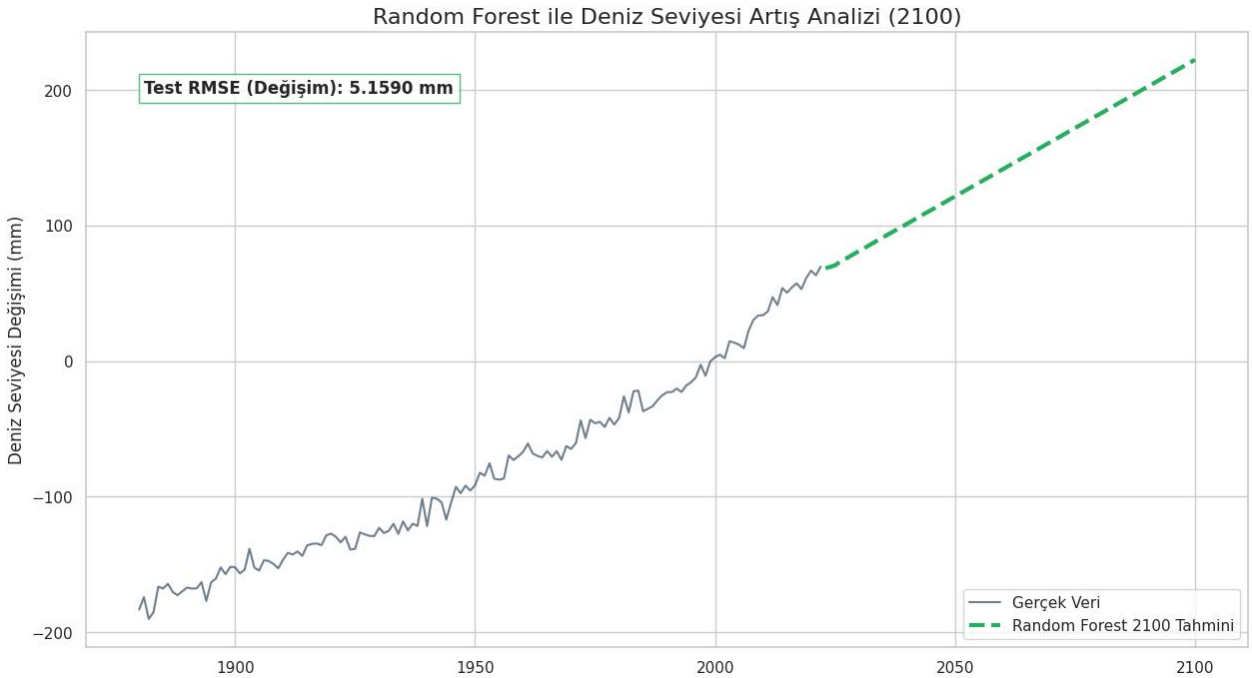


- Test size: 0.05

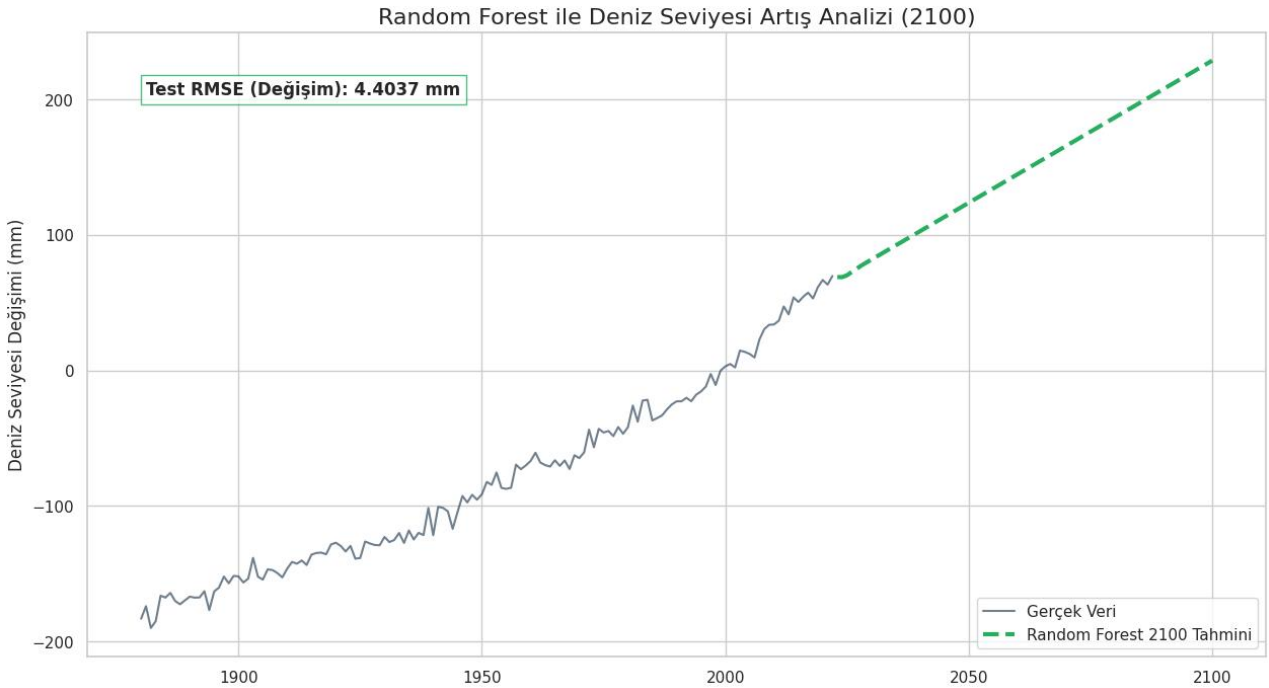


8.2.3 Random Forest:

- Test size: 0.1



- Test size: 0.05



9. Results of Hyperparameter Optimization Experiments:

- For global land temperature forecasting, we achieve the minimum RMSE (0.1495 degrees Celsius) in the final state using 5th-degree polynomial regression with a test size of 0.05.
- For sea level rise, we achieve the minimum RMSE (4.2106 mm) in the final state using the XGBoost algorithm with a test size of 0.05.

Notes: Comparative analyses were conducted across various test sizes, specifically 0.05, 0.10, and 0.20. The results show that the RMSE values were optimized at a 0.05 test split for both datasets. Although a 0.05 test size is relatively small, it consistently yielded the highest predictive accuracy in this specific experimental setup compared to the 0.10 and 0.20 splits. This suggests that the models benefit significantly from a larger training pool. However, the potential for higher variance in performance metrics due to the small test size should be noted as a characteristic of this configuration.

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